MGroups

Finite Group Theory in Mathematica

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MGroups *is a Mathematica Package that implements a part of finite Group Theory. It facilitates studying group operations, order and inverses of the group elements, Cayley tables, (ordinary and normal) subgroup structures of a group, group morphisms and group actions easily.* ***MGroups*** *is licensed under the MIT Open-Source License.*

1. Preface

In **MGroups,** groups are defined in the form of "Associations" that are the Mathematica version of mappings. Quite literally, a group in this package is nothing but its Cayley Table. For example, to define the Z_2 group, all that is needed is to define

1. How 0 operates with 0,
2. How 0 operates with 1,
3. How 1 operates with 0, and
4. How 1 operates with 1.

In Mathematica, Associations are of the form

```
In[1]:= <|a → b, c → d, e → f|>;
```

which means that *a* maps to *b, c* maps to *d,* and *e* maps to *f*. Now, it is only appropriate to see how Z_2 will be defined in such a way:—

```
In[2]:= Z₂ = <|0 → <|0 → 0, 1 → 1|>, 1 → <|0 → 1, 1 → 0|>|>;
```

... which is nothing but nested Associations. Now, if we need to see what is 0*0 in Z_2, we can type

```
In[3]:= Z₂[0][0]
Out[3]= 0
```

to get the required output. Similarly, other complex groups are also defined.

2. Installation

To install the package, you can head over to my GitHub page (`github.com/zplus11/MGroups`) and download the `MGroups.m` file from there. In any Mathematica Notebook, type

```
In[4]:= $UserBaseDirectory <> "\\Applications" (* run this to get your directory *)
Out[4]= C:\Users\Naman Taggar\AppData\Roaming\Mathematica\Applications
```

which should give you a folder path. Place **MGroups.m** file into this path on your device. Then, whenever you need to use **MGroups** in a Mathematica notebook, type

```
In[5]:= << MGroups`
```

which will call the package. Now type

```
In[6]:= AdditiveGroup[2]
Out[6]= ⟨|0 → ⟨|0 → 0, 1 → 1|⟩, 1 → ⟨|0 → 1, 1 → 0|⟩|⟩
```

If you get the same output as above, then congratulations — you have successfully installed **MGroups!**

3. Introduction

This documentation covers each aspect of the package, from basic to advanced. We define a group first:—

Definition *(Group)***.** A group is a non-empty set in Mathematics, elements of which follow 4 properties namely Closure, Associativity, Existence of Identity, and Existence of Inverses under a certain binary operation.

Available Groups

This package provides the groups as tabulated below.

Group	Description
AdditiveGroup	Z_n : The group {0, 1, ..., n – 1} formed under addition modulo n.
MultiplicativeGroup	U_n : The group {0 ≤ x < n : gcd (x, n) == 1} formed under multiplication modulo n.
DihedralGroupp	D_n : The group of symmetries of a regular polygon formed under their composition.
PermutationsGroup	S_A : The group of permutations of a non – empty set formed under their composition.
K_4 & Q_8	Klein's 4 Group and the Quaternion Group.
ExternalDirectProduct	The External Direct Product of given groups.

They can be called by their respective function names.

Defining a Group

Groups can be freely formed with the following command:—

```
In[7]:= domain := {0}
        function[x_, y_] := x + y
        FormGroup[domain, function]
Out[9]= ⟨|0 → ⟨|0 → 0|⟩|⟩
```

without having to type out the Associations yourself. Here, the first argument {0} is the domain, and the second argument is the binary operator that takes two arguments and returns their sum. The resulting group is just the singleton zero {0} with 0 + 0 = 0.

Basic Operations

Import the package by running

In[10]:=
```
<< MGroups`
```

Define a group using one of the functions as shown below.

In[11]:=
```
D4 = DihedralGroupp[4]; (* Caution: DihedralGroup is an inbuilt function,
so this package uses DihedralGroupp *)
```

and check the domain of this group:—

In[12]:=
```
FindDomain[D4]
```

Out[12]=
```
{r0, r1, r2, r3, s0, s1, s2, s3}
```

Group operations can be applied just by indexing those elements through the associations.

In[13]:=
```
D4["r2"]["s3"]
```

Out[13]=
```
s1
```

In[14]:=
```
D4["s1"]["s1"]
```

Out[14]=
```
r0
```

Trying some other group:—

In[15]:=
```
MultiplicativeGroup[13][3][4]
```

Out[15]=
```
12
```

which is 3 × 4 (mod 13). Doing something more exciting...

In[16]:=
```
DihedralGroupp[240]["s60"]["r190"]
```

Out[16]=
```
s10
```

which is the composition of the 191st rotation and reflection about the 61st axis, in the Dihedral group of order 480.

External Direct Products can be formed as follows:—

In[17]:=
```
edp1 =
  ExternalDirectProduct[AdditiveGroup[12], MultiplicativeGroup[20], QuaternionGroup];
```

The order of this group will be 12 × 8 × 8. Let's confirm that.

In[18]:=
```
OrderGroup[edp1]
```

Out[18]=
```
768
```

That is perfect! In EDPs, operations are done component-wise. For example,

In[19]:=
```
el1 = {5, 3, "-j"};
el2 = {8, 19, "k"};
edp1[el1][el2]
```

Out[21]= {1, 17, i}

Group Properties

You can check whether a group is abelian or cyclic using **AbelianQ** or **CyclicQ** commands respectively.

In[22]:=
```
AbelianQ@AdditiveGroup[10]
```

Out[22]= True

In[23]:=
```
AbelianQ@Klein4Group
```

Out[23]= True

In[24]:=
```
CyclicQ@DihedralGroupp[3]
```

Out[24]= False

In[25]:=
```
CyclicQ@ExternalDirectProduct[
  AdditiveGroup[6],
  AdditiveGroup[7]
 ]
```

Out[25]= False

Cayley Tables

Complete Cayley Tables can be printed for any group using **CayleyTable** command.

In[26]:=
```
CayleyTable[QuaternionGroup]
```

Out[26]//TableForm=

	1	-1	i	-i	j	-j	k	-k
1	1	-1	i	-i	j	-j	k	-k
-1	-1	1	-i	i	-j	j	-k	k
i	i	-i	-1	1	k	-k	-j	j
-i	-i	i	1	-1	-k	k	j	-j
j	j	-j	-k	k	-1	1	i	-i
-j	-j	j	k	-k	1	-1	-i	i
k	k	-k	j	-j	-i	i	-1	1
-k	-k	k	-j	j	i	-i	1	-1

or for an EDP:—

In[27]:=
```
CayleyTable@ExternalDirectProduct[AdditiveGroup[2], MultiplicativeGroup[3]]
```

Out[27]//TableForm=

	{0, 1}	{0, 2}	{1, 1}	{1, 2}
{0, 1}	{0, 1}	{0, 2}	{1, 1}	{1, 2}
{0, 2}	{0, 2}	{0, 1}	{1, 2}	{1, 1}
{1, 1}	{1, 1}	{1, 2}	{0, 1}	{0, 2}
{1, 2}	{1, 2}	{1, 1}	{0, 2}	{0, 1}

Orders and Inverses

Table of Orders and Inverses for a group can also be printed as follows:—

In[28]:=
```
InversesTable@DihedralGroupp[6]
```

Out[28]//TableForm=

x	x^{-1}	|x|
r0	r0	1
r1	r5	6
r2	r4	3
r3	r3	2
r4	r2	3
r5	r1	6
s0	s0	2
s1	s1	2
s2	s2	2
s3	s3	2
s4	s4	2
s5	s5	2

In[29]:=
```
ElementInverse[AdditiveGroup[4], 2]
```

Out[29]= 2

In[30]:=
```
InversesTable@PermutationsGroup[{1, 2, 3}]
```

Out[30]//TableForm=

x	x^{-1}	|x|
{1, 2, 3}	{1, 2, 3}	1
{1, 3, 2}	{1, 3, 2}	2
{2, 1, 3}	{2, 1, 3}	2
{2, 3, 1}	{3, 1, 2}	3
{3, 1, 2}	{2, 3, 1}	3
{3, 2, 1}	{3, 2, 1}	2

4. Subgroups

We define a subgroup.

Definition *(Subgroup of a group)***.** A subset $H \subseteq G$ is said to be a subgroup of the group G if it forms a group itself under the operation of G.

We can check this in **MGroups** as follows:—

In[31]:=
```
Z20 = AdditiveGroup[20]; (* {0, 1, ..., 19} *)
sub = Table[2 i, {i, 0, 9}]; (* {0, 2, ..., 18} *)
SubgroupQ[Z20, sub]
```

Out[33]= True

In[34]:=
```
SubgroupQ[DihedralGroupp[10], {"s3", "r0", "r2", "r3"}]
```

Out[34]= False

The packages uses the Finite Subgroup Test to check subgroups. All subgroups can be obtained using

In[35]:=
```
TableForm[Subgroups@DihedralGroupp[5], TableDepth → 1]
```

Out[35]//TableForm=

```
{r0}
{r0, s0}
{r0, s1}
{r0, s2}
{r0, s3}
{r0, s4}
{r0, r1, r2, r3, r4}
{r0, r1, r2, r3, r4, s0, s1, s2, s3, s4}
```

List the cyclic subgroups of U_{30}:—

In[36]:=
```
u30s = Subgroups@MultiplicativeGroup[30];
Keys /@ Select[FormGroup[#, Function[{x, y}, Mod[x y, 30]]] & /@ u30s, CyclicQ@# &]
```

Out[37]= { }

In the case of Cyclic subgroups, finding subgroups is easier (by virtue of Fundamental Theorem of Cyclic Groups). For example, even finding subgroups of U_{997} (order 996) is a doable task.

In[38]:=
```
Subgroups@MultiplicativeGroup[997];
Length[%]
```

Out[39]= 12

5. Subgroup Lattices

Most interestingly, you can also see subgroup lattices of groups using this package. For example, let us see the subgroup lattice of Z_4.

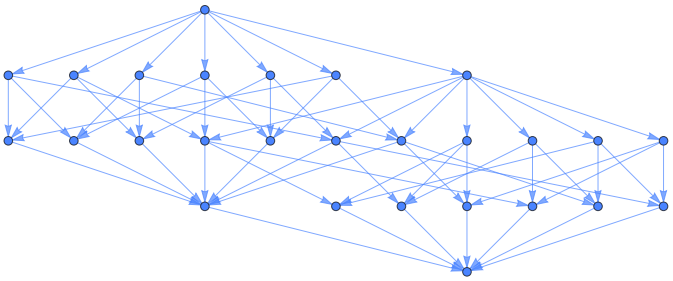
In[40]:=
```
SubgroupLattice@AdditiveGroup[4]
```

Out[40]=

Here, the top most node is the group Z_4 itself, at the bottom we have {0}, and in the middle we must have {0, 2}. Let us proceed further and see something more exciting:—

In[41]:=
```
SubgroupLattice@MultiplicativeGroup[40]
```

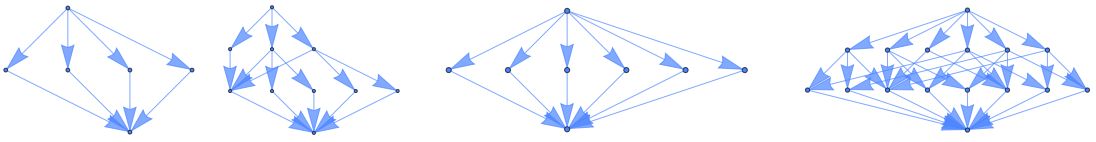
Out[41]=



Beautiful! Subgroup Lattices of some Dihedral Groups:—

In[42]:=
```
GraphicsRow@Table[SubgroupLattice[DihedralGroupp[i]], {i, 3, 6}]
```

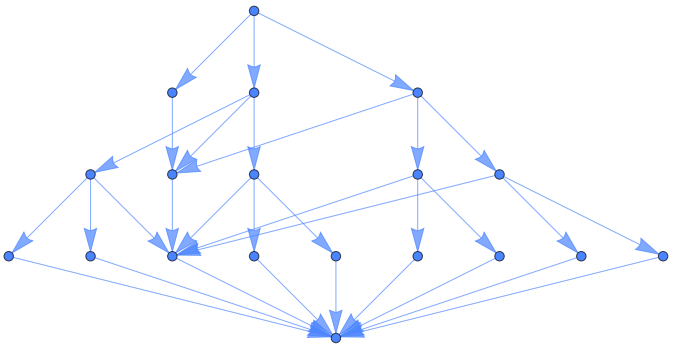
Out[42]=



or D_8:—

In[43]:=
```
SubgroupLattice@DihedralGroupp[8]
```

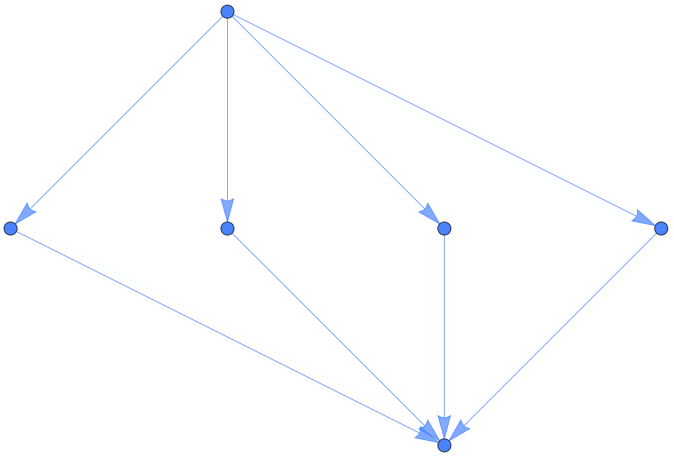
Out[43]=



Symmetric group $S_{\{1,2,3\}}$:—

In[44]:=
```
SubgroupLattice@PermutationsGroup[{1, 2, 3}]
```

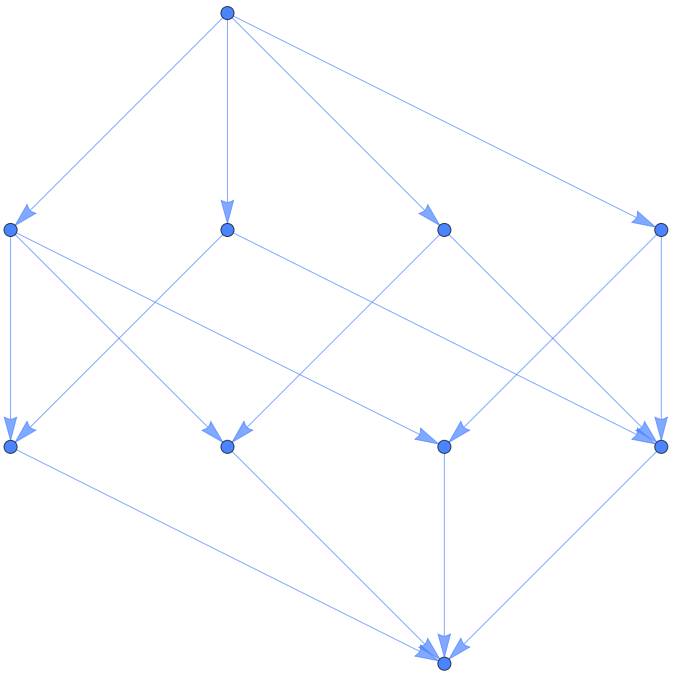
Out[44]=



Some EDPs

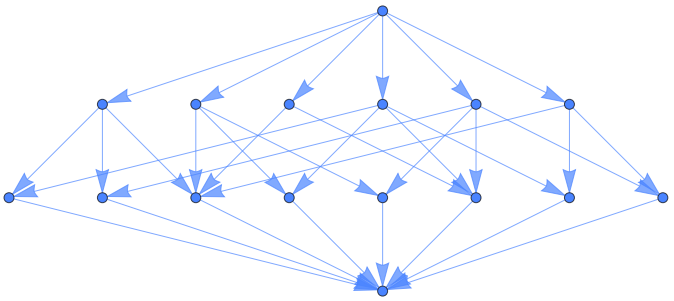
In[45]:=
```
SubgroupLattice@ExternalDirectProduct[AdditiveGroup[6], AdditiveGroup[2]]
```

Out[45]=



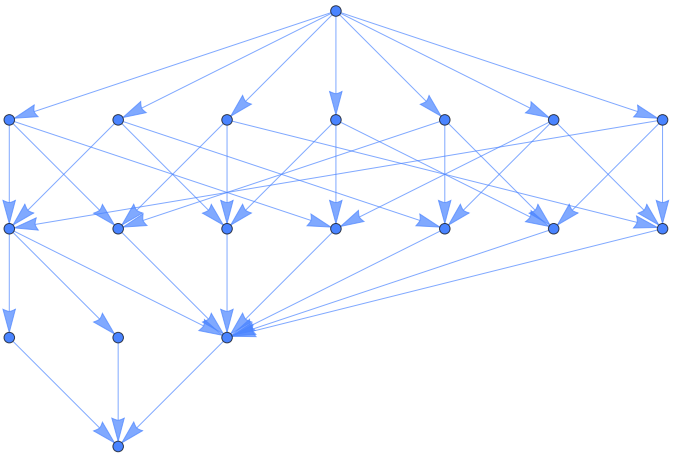
In[46]:=
```
SubgroupLattice@ExternalDirectProduct[DihedralGroupp[3], AdditiveGroup[2]]
```

Out[46]=



In[47]:=
```
ExternalDirectProduct[QuaternionGroup, AdditiveGroup[2]] // SubgroupLattice
```

Out[47]=



6. Cosets and Normal Subgroups

Definition *(Coset)***.** If H is a subgroup of G, then for some a in G, the set $\{a\,h : h \in H\}$ is called the left coset of H in G containing a. Subsequently, corresponding right coset is the set $\{h\,a : h \in H\}$.

You can define cosets in this package as follows:—

In[48]:=
```
edp2 = ExternalDirectProduct[MultiplicativeGroup[8], Klein4Group];
s = {{1, "e"}, {1, "c"}};(* any one subgroup *)
```

In[50]:=
```
Coset[edp2, s, {3, "b"}, "l"]
```

Out[50]=
```
{{3, b}, {3, a}}
```

In[51]:=
```
Coset[edp2, s, {3, "b"}, "r"]
```

Out[51]=
```
{{3, b}, {3, a}}
```

Both are equal.

Definition *(Normal Subgroup of a group)***.** A subgroup H of G is said to be normal in G if the left coset by every element is equal to the right counterpart.

To find normal subgroups of a group:—

In[52]:=
```
TableForm[NormalSubgroups@QuaternionGroup, TableDepth → 1]
```

Out[52]//TableForm=
```
{1}
{1, -1}
{1, -1, i, -i}
{1, -1, j, -j}
{1, -1, k, -k}
{1, -1, i, -i, j, -j, k, -k}
```

7. Morphisms

Definition *(Group Homomorphism)***.** A map ϕ from G_1 to G_2 is said to be a homomorphism, if and only if it is operation preserving, *i.e.,* if $\phi\,(\mathrm{xy}) = \phi\,(x)\,\phi(y)\;\forall\; x,\; y \in G_1$.

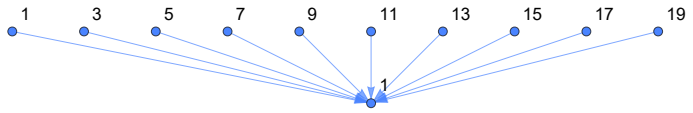
To define a Group Homomorphism, you need to define the domain, the codomain, and the map's definition. It can be done as follows:—

In[53]:=
```
phi = Homomorphism[
  AdditiveGroup[20], (* domain *)
  AdditiveGroup[2], (* co-domain *)
  Mod[#, 2] & (* definition in the form of a function *)
 ]
```

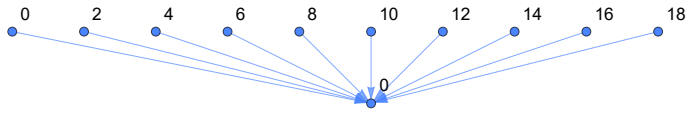
Out[53]=
```
⟨| 0 → 0, 1 → 1, 2 → 0, 3 → 1, 4 → 0, 5 → 1, 6 → 0, 7 → 1, 8 → 0, 9 → 1,
 10 → 0, 11 → 1, 12 → 0, 13 → 1, 14 → 0, 15 → 1, 16 → 0, 17 → 1, 18 → 0, 19 → 1 |⟩
```

Morphisms can be visualised as follows.

In[54]:=
```
VisualiseMorphism@phi
```



Out[54]=



We can see that the kernel of phi is {0, 2, ..., 18}.

Definition *(Group Isomorphism)***.** A homomorphism ϕ is said to be an isomorphism, if and only if it is one-one.

In[55]:=
```
Isomorphism[
 AdditiveGroup[3],
 AdditiveGroup[3],
 # &
]
```

Out[55]= ⟨| 0 → 0, 1 → 1, 2 → 2 |⟩

Definition *(Group Automorphism)***.** An isomorphism from a group to itself is said to be an automorphism.

In[56]:=
```
Automorphism[
 AdditiveGroup[3],
 # &
]
```

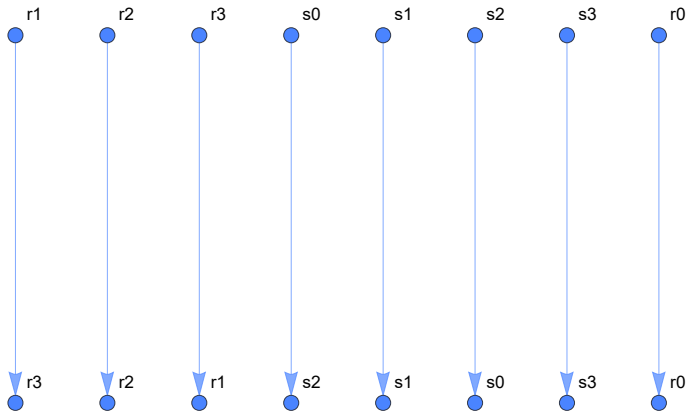
Out[56]= ⟨| 0 → 0, 1 → 1, 2 → 2 |⟩

Definition *(Inner Automorphism)***.** An inner automorphism is an automorphism that is induced by a group element a, with the definition $x \longrightarrow a\, x\, a^{-1}$.

In[57]:=
```
InnerAutomorphism[DihedralGroupp[4], "s3"]
VisualiseMorphism[%]
```

Out[57]= ⟨| r0 → r0, r1 → r3, r2 → r2, r3 → r1, s0 → s2, s1 → s1, s2 → s0, s3 → s3 |⟩

Out[58]=



Work to be continued on this section...

8. Group Actions

Definition *(Group Action)*. A group action of a group G on a set A is a map from $G \times A$ to A defined by $(g, a) \to g.a \; \forall \; g \in G, \; a \in A$ that satisfies the following properties:

1. $g_1.(g_2.a) = (g_1 \, g_2).a \; \forall \; g_1, \; g_2 \in G, \; a \in A$, and
2. $e.a = a \; \forall \; a \in A$

where e is the identity of G.

We can define a group action as follows:

In[59]:=
```
GroupAction[
 PermutationsGroup[{1, 2, 3}],
 {1, 2, 3},
 #1[[#2]] &
]
```

Out[59]=
```
⟨| {{1, 2, 3}, 1} → 1, {{1, 2, 3}, 2} → 2, {{1, 2, 3}, 3} → 3,
 {{1, 3, 2}, 1} → 1, {{1, 3, 2}, 2} → 3, {{1, 3, 2}, 3} → 2, {{2, 1, 3}, 1} → 2,
 {{2, 1, 3}, 2} → 1, {{2, 1, 3}, 3} → 3, {{2, 3, 1}, 1} → 2, {{2, 3, 1}, 2} → 3,
 {{2, 3, 1}, 3} → 1, {{3, 1, 2}, 1} → 3, {{3, 1, 2}, 2} → 1, {{3, 1, 2}, 3} → 2,
 {{3, 2, 1}, 1} → 3, {{3, 2, 1}, 2} → 2, {{3, 2, 1}, 3} → 1 |⟩
```

Definition *(Permutation Representation)*. The map from G to S_A defined by $g \to \sigma_g$ is a homomorphism and is called the permutation representation associated with the given group action, where:—

$\sigma_g : A \to A$ defined by $\sigma_g(a) = g.a$ is a permutation on the set A called the permutation induced by a, for each $a \in A$.

The associated permutation representation to this action is nothing but the identity map from S_A to itself, as shown below:—

In[60]:=
```
PermutationRepresentation[
 PermutationsGroup[{1, 2, 3}],
 {1, 2, 3},
 #1[[#2]] &
]
```

Out[60]=
```
⟨| {1, 2, 3} → {1, 2, 3}, {1, 3, 2} → {1, 3, 2}, {2, 1, 3} → {2, 1, 3},
 {2, 3, 1} → {2, 3, 1}, {3, 1, 2} → {3, 1, 2}, {3, 2, 1} → {3, 2, 1} |⟩
```

A. Copyright

Github Repository: `https://github.com/zplus11/MGroups.git`.